Value investing: buy-and-hold results

S&P 500, data as of 2026-10-31

Abstract

Each January and July since 2016, a rule-based value score picks 10 S&P 500 companies, and 100 USD is invested in them, weighted by score and never sold. Investors who started in 2016, 2020, 2023 and 2025 (an earlier start buys from 2016) are followed to 30 October 2026, each against the same money in the S&P 500 total-return index, and in Berkshire Hathaway. The reference is the investor who started in 2020: the data hold only today's index members, and by the index's list of changes, which starts in July 1976, 125 companies have left the index since then against 230 since 2016, so fewer failed companies are missing. That investor's 1,400 USD is worth 3,947 USD, against 2,480 USD in the benchmark and 2,415 USD in Berkshire Hathaway; a second wallet that also sells holdings whose score has fallen is worth 3,927 USD. The report closes with the next picks: First Solar (FSLR), Snap-on (SNA), Cognizant (CTSH), ResMed (RMD), Edwards Lifesciences (EW), Skyworks Solutions (SWKS), BrownForman (BF-B), ON Semiconductor (ON), Insulet Corporation (PODD) and Fastenal (FAST). The result rests on few decisions and few companies, with 33% of the value in one company; it is a research result, not evidence of a lasting edge.

1 Method

Company fundamentals are the values first reported in each annual report (10-K) to the SEC, used from the day they were filed. When a company reports an item under more than one label, the label it used most recently by the date is taken, so later filings never decide it (Appendix D). Prices and dividends are from Yahoo Finance; returns include reinvested dividends and are in US dollars. On each date, only companies already in the S&P 500 are considered, outside the Financials and Real Estate sectors.

Ten rules from Graham and Buffett grade every company: G6 and G7 (price against earnings and book value), G2a and E1 (current ratio), G2b and E2 (debt), NN (net-net), G5 (earnings growth), B1 (return on equity) and E4 (current dividend). Six more (G3, G4, E3, E5, E6 and NI1) are switched off and can be switched on. For each rule, the companies are ranked by the rule's value and placed in six equal groups, from 5 points for the best sixth to 0 for the worst; each rule's points are multiplied by its weight, one less the share of companies that pass the rule's threshold, so rules few companies pass count more. The score is the sum. On the first trading day of January and of July, the 10 highest-scoring companies are bought for 100 USD, each in proportion to its score.

Three limits apply to every number in this report. Companies that left the index are missing from the data, which flatters the results. The score's design and every setting of this report (the rules and their weighting, the number of picks, the decision months, the sell threshold, the start years) were chosen while looking at the whole period from 2016, so no period here is out of sample: the results describe how these rules would have done, not evidence that they will do so. There are no trading costs or taxes.

2 Picks per half-year

Table 1: Purchases per half-year and their value today (USD).

Bought	Picks and weights	Wallet	S&P 500	Berkshire
2016-01	CMI 11%, WDC 11%, CSCO 11%, QCOM 10%, TPR 10%, GWW 10%, NTAP 10%, GLW 9%, HAL 9%, PFE 9%	621	452	415
2016-07	CSCO 11%, WDC 11%, QCOM 11%, NTAP 10%, TPR 10%, CMI 10%, CTSH 10%, VLO 9%, TEL 9%, BBY 9%	633	428	377
2017-01	CSCO 12%, CTSH 11%, LRCX 10%, QCOM 10%, NKE 10%, VLO 10%, CMI 10%, AMGN 9%, GILD 9%, FAST 9%	616	394	331
2017-07	CSCO 11%, QCOM 10%, CTSH 10%, FAST 10%, TSCO 10%, BBY 10%, GLW 10%, BIIB 9%, VLO 9%, LRCX 9%	539	363	316
2018-01	CSCO 11%, CTSH 11%, LRCX 10%, TPR 10%, GLW 10%, QCOM 10%, BBY 10%, AMAT 10%, TSCO 10%, VLO 9%	483	324	275
2018-07	QCOM 11%, LRCX 10%, CSCO 10%, AMAT 10%, LYB 10%, GILD 10%, FAST 10%, TPR 10%, CTSH 10%, SNA 9%	485	317	289
2019-01	LRCX 12%, MU 12%, KLAC 10%, LYB 10%, CTSH 10%, AMAT 10%, GILD 10%, FAST 9%, VLO 9%, HAS 9%	1,060	341	267
2019-07	MU 12%, NUE 11%, LRCX 11%, SNA 10%, CTSH 10%, NVDA 10%, FAST 9%, KLAC 9%, HUM 9%, AMAT 9%	1,260	286	253
2020-01	MU 12%, NUE 12%, CTSH 11%, SNA 10%, FAST 9%, TPR 9%, AMAT 9%, PKG 9%, GILD 9%, LYB 9%	508	257	238
2020-07	MU 12%, NUE 11%, CTSH 11%, HAS 10%, HUM 10%, PKG 10%, SNA 9%, FAST 9%, TPR 9%, HSIC 9%	538	266	305
2021-01	MU 11%, NUE 11%, HUM 10%, CTSH 10%, AMAT 10%, HAS 10%, PKG 10%, LRCX 9%, GILD 9%, BIIB 9%	396	223	237
2021-07	SNA 12%, MU 11%, HUM 10%, CMI 10%, INTC 10%, FAST 10%, NUE 10%, BBY 9%, HSIC 9%, CTSH 9%	274	189	194
2022-01	SNA 11%, MU 11%, HUM 10%, CMI 10%, HSIC 10%, INTC 10%, NUE 10%, AMAT 10%, PKG 9%, CTSH 9%	259	169	180
2022-07	SNA 11%, NUE 11%, MU 11%, AMAT 10%, SWKS 10%, GOOG 10%, GOOGL 10%, CMI 10%, INTC 10%, CTSH 9%	393	211	195
2023-01	MU 11%, SNA 10%, GOOG 10%, GOOGL 10%, NUE 10%, CTSH 10%, LRCX 10%, SWKS 10%, CMI 9%, INTC 9%	419	209	175
2023-07	STLD 12%, NUE 11%, SNA 11%, MU 10%, CF 10%, DD 10%, CTSH 10%, LRCX 9%, SWKS 9%, GOOG 9%	313	178	159
2024-01	STLD 11%, NUE 11%, SWKS 11%, SNA 11%, CF 10%, DD 10%, CTSH 9%, AMAT 9%, KEYS 9%, LRCX 9%	177	166	150
2024-07	NUE 11%, SNA 11%, STLD 10%, BG 10%, SWKS 10%, EOG 10%, CTSH 10%, CF 9%, ON 9%, EBAY 9%	147	142	134
2025-01	NUE 11%, SWKS 10%, STLD 10%, BG 10%, SNA 10%, EOG 10%, CTSH 10%, AMAT 9%, ON 9%, CF 9%	157	132	120
2025-07	ON 11%, SWKS 11%, SNA 11%, EW 10%, BG 10%, FSLR 10%, CTSH 10%, AMAT 9%, STLD 9%, EOG 9%	135	124	111
2026-01	ON 11%, SNA 11%, SWKS 10%, EW 10%, CPRT 10%, CTSH 10%, BG 10%, EOG 10%, NUE 10%, STLD 9%	121	112	109
2026-07	SNA 11%, CTSH 11%, FSLR 11%, SWKS 10%, RMD 10%, EW 10%, BF-B 10%, PODD 9%, CF 9%, HUM 9%	110	102	109

Table 1 lists the companies bought each January and July, with their weights, and what each 100 USD is worth on 30 October 2026, next to the same money in the S&P 500 and in Berkshire Hathaway. Figure 1 shows the same values. Most of the money went to Information Technology, Industrials and Materials; the best single purchase is NVDA in July 2019, now worth 50 times its cost. The purchases of none are worth less today than the same money in the S&P 500. Appendix

A lists every purchase of the wallet with sells of the investor who started in 2020, with its price at purchase and today, what later sales took from it, what it keeps today and its total return with dividends.

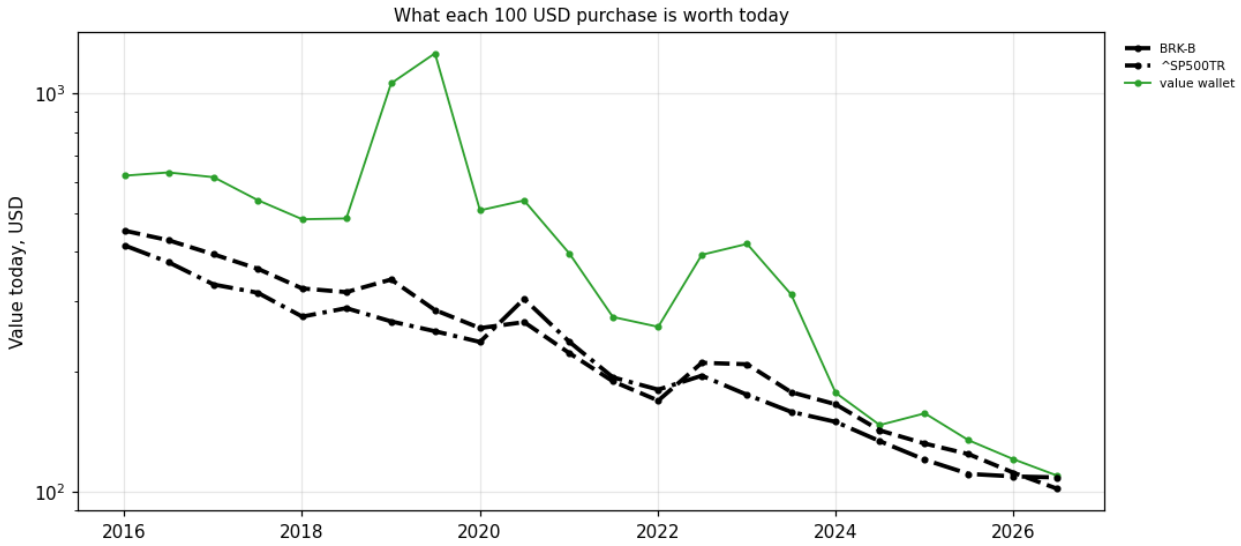


Figure 1: What each purchase is worth today, by date of purchase, on a logarithmic scale.

3 Value over time

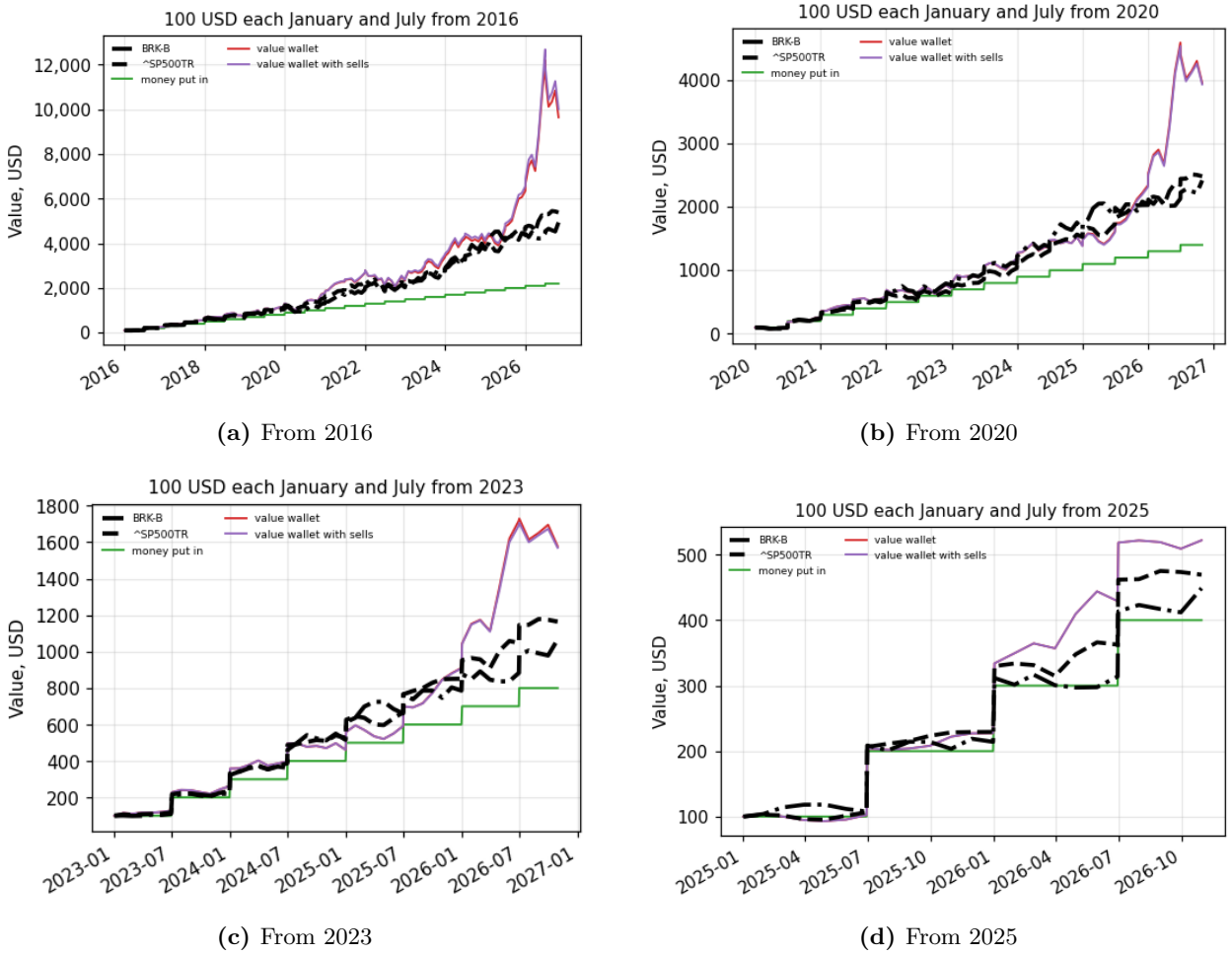
Table 2: Money put in and value today per start year (USD), and the money-weighted return a year.

Start	Holder	Put in	Value	Multiple	A year
2016	Value wallet	2,200	9,644	4.38	24.9%
	Value wallet with sells	2,200	10,014	4.55	25.5%
	S&P 500	2,200	5,383	2.45	15.3%
	Berkshire Hathaway	2,200	4,938	2.24	13.9%
2020	Value wallet	1,400	3,947	2.82	28.9%
	Value wallet with sells	1,400	3,927	2.80	28.8%
	S&P 500	1,400	2,480	1.77	15.9%
	Berkshire Hathaway	1,400	2,415	1.73	15.2%
2023	Value wallet	800	1,577	1.97	34.9%
	Value wallet with sells	800	1,569	1.96	34.6%
	S&P 500	800	1,164	1.46	18.7%
	Berkshire Hathaway	800	1,066	1.33	14.2%
2025	Value wallet	400	522	1.30	27.0%
	Value wallet with sells	400	522	1.30	27.0%
	S&P 500	400	469	1.17	15.7%
	Berkshire Hathaway	400	449	1.12	11.1%

Table 2 and Figure 2 follow the four investors. Each puts in 100 USD every January and July from the start year; the value wallet never sells. The return a year is money-weighted: the constant yearly rate that turns the payments into today's value.

The investor who started in 2016 earns 24.9% a year against 15.3% in the S&P 500 and 13.9% in Berkshire Hathaway. From 2020 the rates are 28.9%, 15.9% and 15.2%; from 2023 34.9%, 18.7% and 14.2%; from 2025, over less than two years, 27.0%, 15.7% and 11.1%. The wallet's value swings more than either comparison, because 10 companies of a few industries are far less diversified than the index or Berkshire.

The value wallet with sells buys the same picks with the same money, but at each decision it also sells part of any holding whose score has fallen; Section 4 describes the rule. From 2016 it is worth 10,014 USD (25.5% a year), from 2020 3,927 USD (28.8%) and from 2025 522 USD (27.0%).



(a) From 2016

(b) From 2020

(c) From 2023

(d) From 2025

Figure 2: Value of 100 USD every January and July from each start year, in USD, next to the money put in; the S&P 500 dashed and Berkshire Hathaway dash-dot.

4 Selling below the threshold

Each January and July, before buying, the wallet with sells checks every company it holds. The score is put on a scale from 0 to 1, the share of the most the company could score on the rules it is measured on. A holding scoring 0.4 or less is sold in part: the share sold is (0.4 less the score) divided by 0.4, so all of it at a score of 0 and almost none just under 0.4. Holdings are sold, never sold short. The proceeds are added to the decision's 100 USD and buy its picks.

4.1 Sold in the wallet

Table 3: Sales of the investor who started in 2020: share of each holding sold, its score on the 0-to-1 scale, and the proceeds.

Year	Sold: share of the holding (score)	Proceeds, USD
2021-01	TPR 11% (score 0.36)	3.64
2021-07	TPR 23% (score 0.31), HAS 0% (score 0.40)	9.33
2022-01	HAS 2% (score 0.39)	0.53
2023-01	GILD 16% (score 0.34)	4.49
2023-07	GILD 1% (score 0.40), HAS 23% (score 0.31)	4.00
2024-01	GILD 0% (score 0.40), HAS 10% (score 0.36)	1.07
2024-07	HAS 73% (score 0.11), BBY 11% (score 0.36)	8.95
2025-01	HAS 74% (score 0.11), CMI 1% (score 0.39), BBY 11% (score 0.36)	3.89
2025-07	GILD 1% (score 0.40), HAS 20% (score 0.32), INTC 46% (score 0.22), BBY 5% (score 0.38), EBAY 27% (score 0.29)	15.76
2026-01	GILD 1% (score 0.40), HAS 21% (score 0.32), INTC 48% (score 0.21), BBY 4% (score 0.39), EBAY 20% (score 0.32)	14.15
2026-07	HAS 32% (score 0.27), INTC 13% (score 0.35), EBAY 26% (score 0.30)	8.40

Table 3 lists the 27 sales of the investor who started in 2020, worth 74 USD in all. They are the wallet's own cash flows; the list below does not depend on what anyone holds.

4.2 To sell or short

Table 4: The 10 lowest-scoring companies on the latest score date, as many as the picks: sector, score, score on the 0-to-1 scale and what the score suggests.

Company	Sector	Score	Score 0 to 1	What it suggests
EchoStar (ECHO)	Communication Services	0.8	0.03	start thinking about selling
Palo Alto Networks (PANW)	Information Technology	1.4	0.06	start thinking about selling
GE Vernova (GEV)	Industrials	1.6	0.07	start thinking about selling
J.M. Smucker Company (The) (SJM)	Consumer Staples	1.9	0.05	start thinking about selling
Dell Technologies (DELL)	Information Technology	2.4	0.07	start thinking about selling
Exelon (EXC)	Utilities	3.3	0.13	start thinking about selling
AbbVie (ABBV)	Health Care	3.4	0.09	start thinking about selling
Cardinal Health (CAH)	Health Care	3.6	0.10	start thinking about selling
Hilton Worldwide (HLT)	Consumer Discretionary	3.8	0.10	start thinking about selling
Live Nation Entertainment (LYV)	Communication Services	3.9	0.10	start thinking about selling

These are the 10 lowest scores in the S&P 500 on 18 September 2026, as many as the picks, whether or not anyone holds them: EchoStar (ECHO), Palo Alto Networks (PANW), GE Vernova (GEV), J.M. Smucker Company (The) (SJM), Dell Technologies (DELL), Exelon (EXC), AbbVie (ABBV), Cardinal Health (CAH), Hilton Worldwide (HLT) and Live Nation Entertainment (LYV) (Table 4). The last column reads each score on the 0-to-1 scale against the sell threshold of 0.4. At or below the threshold, the company is one to avoid, and to hold only if you already own it. At or below half

of the threshold, it is one to start thinking about selling. At zero, the score is not worth anything. The cut-offs are those the score uses today and are not fixed. The wallet never sells short; a short position on these companies is a later step and has not been tested.

5 Wallet contents

Figure 3 follows every company in the wallet with sells of the investor who started in 2020, month by month. A line starts or steps up when the company is bought, steps down when part of it is sold, and rises or falls with the price in between. The dashed lines follow the same money in the S&P 500 and in Berkshire Hathaway, divided by the number of companies the wallet holds on the date, so they show what one holding would be worth in the index or in Berkshire. Appendix B gives each holding's value after every January's trades, the money put in and the proceeds of sales.

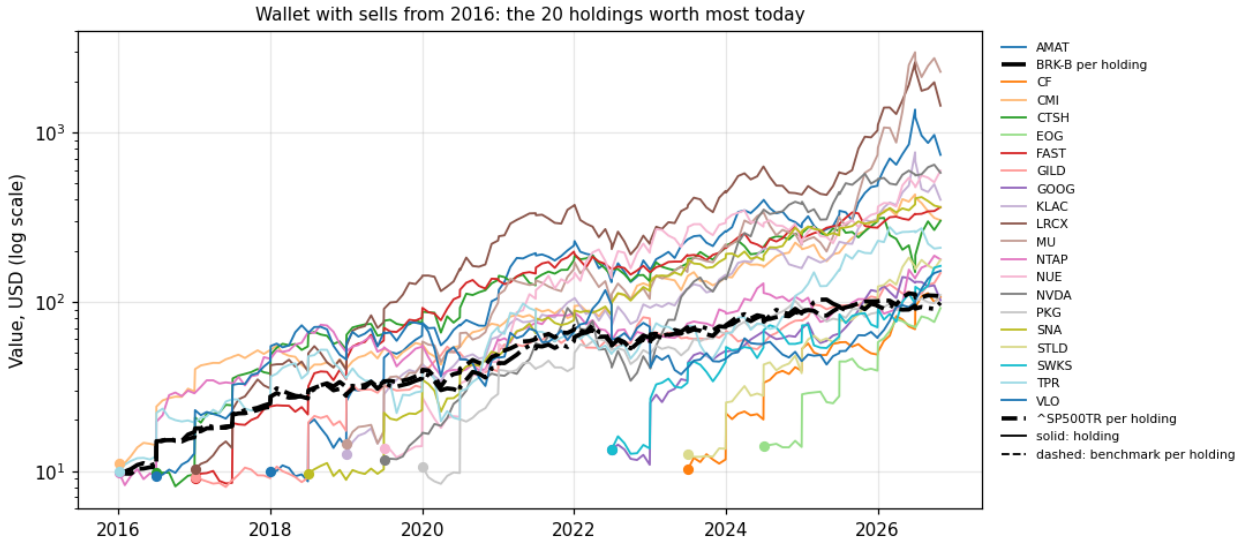


Figure 3: Value of each holding in the wallet with sells (solid) and of the S&P 500 and Berkshire Hathaway divided by the number of holdings (dashed), in USD on a logarithmic scale.

6 Rules behind the winners

Table 5: Dollars ahead of the same money in the S&P 500, per pick: total, average and median of the picks that beat the index and of those that lagged it.

Picks	Number	USD ahead, total	Average	Median
Beat the S&P 500	126	+4,986	+39.6	+9.5
Lagged the S&P 500	94	-725	-7.7	-5.3
All picks	220	+4,261	+19.4	+1.6

Of the picks, 126 have done better than the S&P 500 since their purchase and 94 have not, but a count treats a large win and a small loss alike. Table 5 weighs them in dollars: each pick's value today less what the same money would be worth in the S&P 500. The winners are 4,986 USD ahead of the index and the laggards 725 USD behind, a net 4,261 USD, the wallet's whole lead over the index. The median winner is 9 USD ahead and the median laggard 5 USD behind; the average win is 5.1 times the average loss. The wins are concentrated: the 5 largest, NVDA Jul 2019, MU Jan 2019, MU Jul 2019, LRCX Jan 2017 and LRCX Jan 2019, make up 28% of the winners' total.

Figure 4 compares the mean points per rule of the winners and of the laggards on the day they were picked, each pick weighted by its dollars ahead or behind the index, so the large wins and losses

count most; the rules are ordered by how much more the winners scored. The winners scored most above the others on net-net (NN), 6-year earnings growth (G5), current ratio (G2a) and current ratio (lenient) (E1); the picks that lagged scored higher on return on equity (B1) and current dividend (E4). With so few picks, these differences describe this sample; they do not establish which rules work. Appendix E explains each rule, and Appendix C tests whether the rule weights favour the rules that told winners from losers.

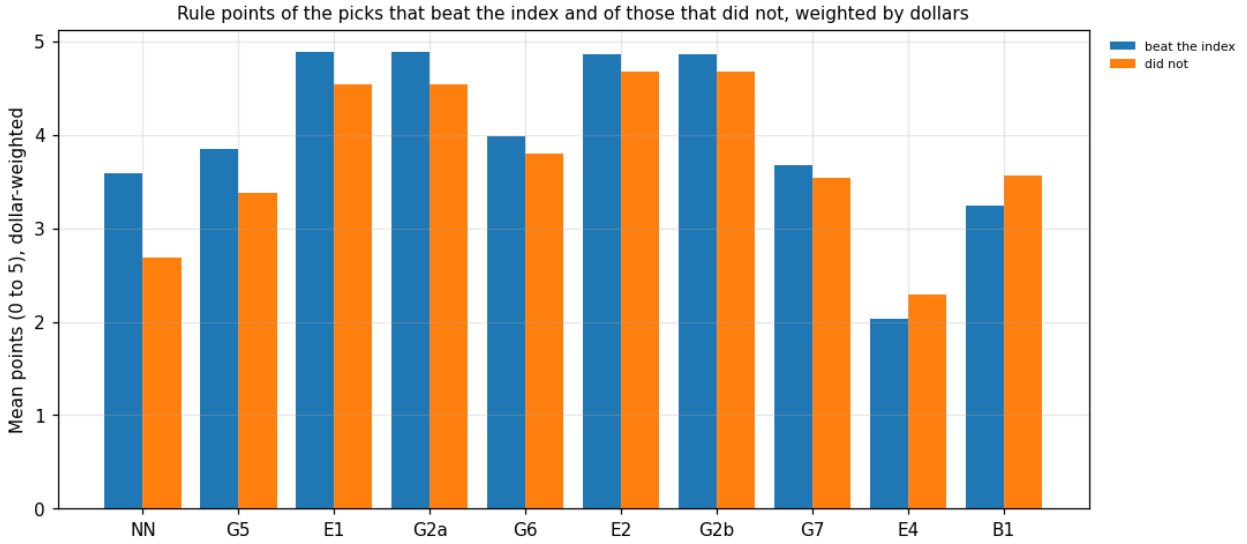


Figure 4: Mean points per rule (0 to 5) of the picks that beat the S&P 500 since purchase and of those that did not, each pick weighted by its dollars ahead or behind the index.

7 The Magnificent 7

Table 6: The Magnificent 7 on the score: score and score on the 0-to-1 scale on the latest score date, rank among the companies scored then, and median rank on the dates of the decisions since 2016.

Company	Score	Score 0 to 1	Rank today	Median rank
Nvidia (NVDA)	22.5	0.60	28 of 379	32
Alphabet Inc. (Class A) (GOOGL)	21.4	0.57	44 of 379	22
Meta Platforms (META)	21.1	0.76	50 of 379	178
Microsoft (MSFT)	19.0	0.51	78 of 379	50
Tesla, Inc. (TSLA)	18.1	0.48	94 of 379	112
Amazon (AMZN)	12.4	0.33	212 of 379	229
Apple Inc. (AAPL)	7.0	0.19	340 of 379	151

The best-known growth companies of the period, the Magnificent 7, are rarely picked. On the latest score date Nvidia (NVDA) ranks highest of them, 28 of 379 companies, and Apple Inc. (AAPL) lowest at 340 (Table 6). The rules are Graham's and Buffett's, written to avoid paying a high price for growth: Figure 5 shows that the Magnificent 7 score far fewer points than the picks on P/E times P/B (G7), P/E (G6), current dividend (E4) and net-net (NN), and more only on 6-year earnings growth (G5).

The same 100 USD at each decision, split equally over the Magnificent 7 in the index on the date and never sold, is worth 23,955 USD from 2016 (39.9% a year), 4,445 USD from 2020 (32.3%) and 462 USD from 2025 (14.1%), against 9,644, 3,947 and 522 USD in the value wallet. The comparison picks the Magnificent 7 with hindsight: they are named for having been the winners of these years.

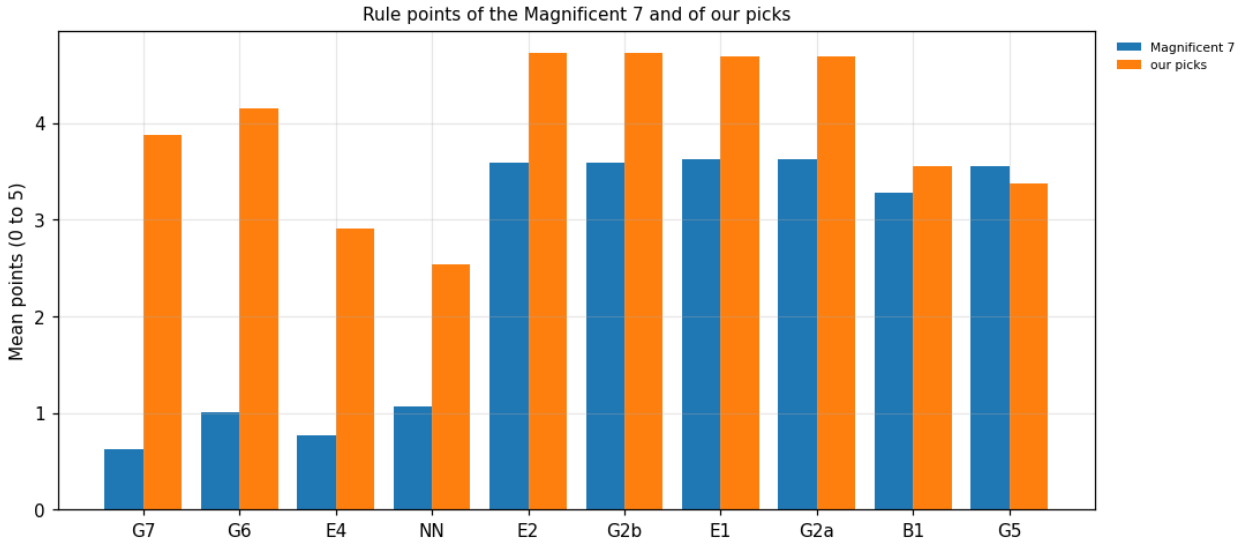


Figure 5: Mean points per rule (0 to 5) of the Magnificent 7 and of our picks on the decision dates since 2016, the rules ordered by how many more points the picks scored.

8 Price history of the picks

Figure 6 shows the share price of every company the investor who started in 2023 bought or sold, and of the next picks, over the last 5 years. The price is split-adjusted and excludes dividends, as a price chart shows it; the companies run from the best to the worst price change over the 5 years, given beside each name; when they do not fit on one page, only the 6 best and the 6 worst are shown. Each purchase is marked in green and each sale in red, at the price on the decision date; a larger marker means more money. Figure 7 gives the same trades in USD per company and decision date.

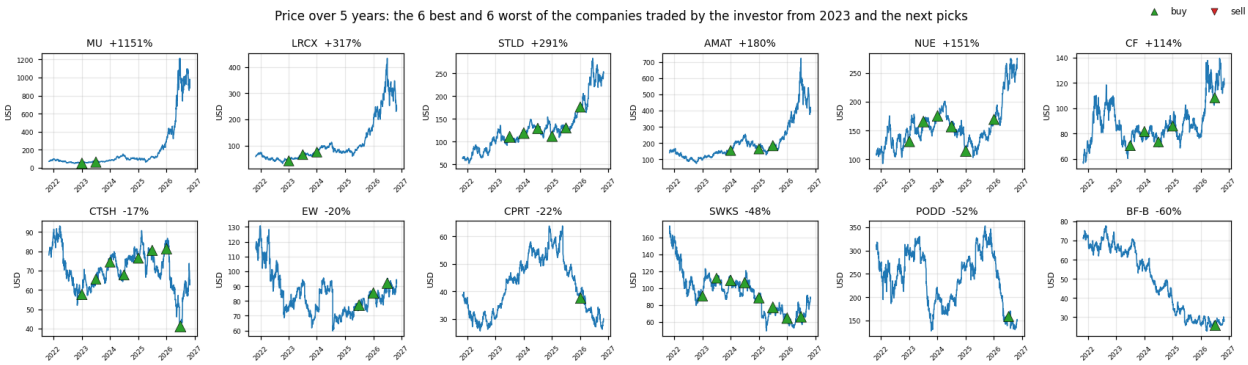


Figure 6: Share price (USD, split-adjusted, dividends excluded) over the last 5 years, one panel per company, ordered from the best to the worst price change over the period; purchases (green) and sales (red) of the investor who started in 2023, marker area in proportion to the USD traded.

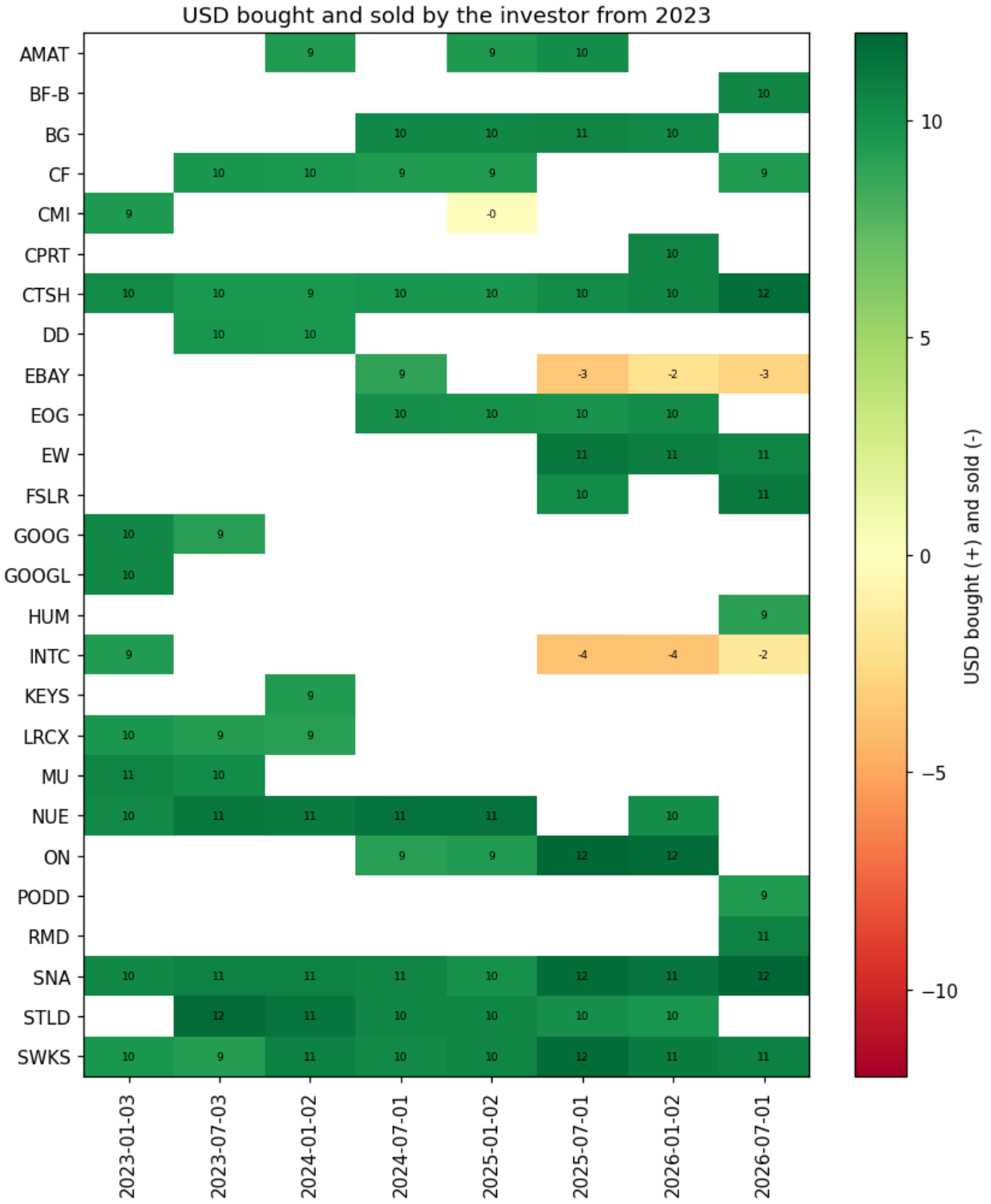


Figure 7: USD bought (green, positive) and sold (red, negative) by the investor who started in 2023, per company and decision date.

9 The next picks

Table 7: The next picks on the latest score date: score, score on the 0-to-1 scale, weight and USD of each 100 USD.

Company	Sector	Score	Score 0 to 1	Weight	USD
First Solar (FSLR)	Technology	34.6	0.93	11.7%	11.71
Snap-on (SNA)	Industrials	33.4	0.89	11.3%	11.29
Cognizant (CTSH)	Technology	31.4	0.84	10.6%	10.63
ResMed (RMD)	Health care	30.7	0.82	10.4%	10.37
Edwards Lifesciences (EW)	Health care	29.4	0.79	9.9%	9.94
Skyworks Solutions (SWKS)	Technology	29.1	0.78	9.8%	9.82
BrownForman (BF-B)	Staples	27.8	0.74	9.4%	9.39
ON Semiconductor (ON)	Technology	27.3	0.73	9.2%	9.23
Insulet Corporation (PODD)	Health care	27.0	0.72	9.1%	9.12
Fastenal (FAST)	Industrials	25.1	0.67	8.5%	8.50

On the latest score date, 18 September 2026, the 10 highest-scoring companies are First Solar (FSLR), Snap-on (SNA), Cognizant (CTSH), ResMed (RMD), Edwards Lifesciences (EW), Skyworks Solutions (SWKS), BrownForman (BF-B), ON Semiconductor (ON), Insulet Corporation (PODD) and Fastenal (FAST) (Table 7). Each would receive its share of the next decision's 100 USD and of any proceeds from sales. The decision is taken on the first trading day of January with the scores known then, which include the annual reports filed until that day; these picks are a preview and may change. Figure 8 shows each pick's share price over the last 5 years, with the earlier purchases of the investor who started in 2023 and the potential purchase at today's price.



Figure 8: Share price of each next pick (USD, split-adjusted, dividends excluded) over the last 5 years: earlier purchases of the investor who started in 2023 in green, the potential purchase at the latest price in orange.

10 Winning picks we missed

Table 8: Winners over the next 36 months that the score missed, by index: the share of winners held (by chance), their mean score and examples with their score on a scale from 0 to 1.

Index	Winners held (by chance)	Winners' mean score	Missed winners and their score
S&P 500	10% (4%)	0.55	Amazon 0.08, Netflix 0.23, Constellation Brands 0.32, Adobe 0.48
S&P 400	5% (9%)	0.42	Paylocity 0.03, Repligen 0.16, Tenet 0.17, Qualys 0.20, Chemed 0.34
S&P 600	9% (5%)	0.51	Archrock 0.14, Brinker 0.16, SM Energy 0.19

No score picks every winner, and a winner is only known afterwards. Here the winners are the ten companies with the highest return above their index over the next 36 months, found with hindsight on every date. A book of ten companies that knew nothing would hold about 4% of them on the S&P 500, 9% on the S&P 400 and 5% on the S&P 600 (Table 8). They are shown as a check on the rules, not as advice.

On the S&P 500 the score holds 10% of the winners and ranks them slightly above the median (0.55 on a scale from 0 to 1). It found the chip makers (Nvidia 0.91, Lam Research 0.96, Micron 0.91) and missed the growth platforms: Amazon 0.08, Netflix 0.23 and Constellation Brands 0.32. On the S&P 400 the winners score below the median (0.42): Paylocity 0.03, Repligen 0.16, Tenet 0.17 and Qualys 0.20. Health care is 17% of the winners there and 3% of our picks. On the S&P 600 energy companies are 24% of the winners and 2% of our picks.

The likely cause is in the rules. They are Graham's and Buffett's, and they reward cheap, low-debt, asset-rich companies. Many winners of the period were growth and quality companies that pay no dividend and trade at higher multiples. On the S&P 400 they score 0.7 points lower than the average company on the dividend rule and 0.4 lower on debt cover, while our picks score 4.4 on debt cover against 2.1 for the average company. Dropping the dividend rules made no measurable difference, so they are not what holds the score back; the price rules are the more likely cause (Section 12). Companies that S&P later added to the S&P 500 scored lower than its existing members before they joined (0.42 against 0.51 on a scale from 0 to 1), which points the same way: the score undervalues growth.

11 How the model was tested

Table 9: The tests the model must pass before launch, and where each stands.

Test	What is checked	Result	Status
Winners removed	The wallet still beats its index with the 3 largest winners removed	S&P 500: yes, by 1.9 to 3.7 points (2016 to 2023 starts). S&P 400: no. S&P 600: mixed	S&P 500 passed; others not
Company samples	12 samples, each leaving out a random tenth of the companies	S&P 500: ahead of the index in 12 of 12	Passed on the S&P 500
Unseen companies	The score frozen, then run once on a tenth of the companies never used	Positive, but two companies gave 99% of the gain (earlier sixteen-rule score)	To rerun
Other indices	The score beats random picks of its own index on the S&P 400 and 600	Not reproduced	Open
Two periods	The edge appears before and after 2017	Yes on the S&P 500 (earlier sixteen-rule score)	To rerun
Missed winners	Which winners the score missed, and why	Section 10	Done
Survivorship	The effect of companies that left the index	Not yet measured	Open
Real-world check	Results compared with market events and institutional logic	Not yet done	Open

The score is judged only on what it did not know: companies and dates that were held out, against random picks of the same size, with the uncertainty stated and the number of designs tried counted. The tests follow a written protocol, and a test that failed is shown as failed. The numbers are from runs on 1 November 2026 on the ten default rules, with data as of 31 October 2026.

On the S&P 500 the picks beat the index in all 12 samples that each left out a random tenth of the companies, by 9 to 15 points a year on average depending on the start year, and by about 5 points in the worst sample. They still beat it with the three largest winners removed as if they had never existed (Applied Materials, Lam Research and Micron): by 3.7, 1.9 and 2.1 points a year for the investors who started in 2016, 2020 and 2023, and by 8.9 for 2025. With five removed (adding Nucor and Nvidia) the margin shrinks to 0.3 to 2.3 points for the first three. The edge is real on this index but thin, and concentrated in chip and materials companies that did well in a long bull market after 2009.

The other two indices do not confirm it. On the S&P 400 the picks beat their fund by 2 to 5 points a year for the 2016 to 2023 starts, and with the three largest winners removed the margin is gone (0.4, -0.4 and -0.1). On the S&P 600 the investors who started in 2016 and 2020 are ahead by about 6 points, which survives the removal of the three largest winners, while those who started in 2023 and 2025 trail the fund. The lists of the S&P 400 and 600 give no entry dates and lose the companies that grew into a larger index, so these two tests are biased and not yet a verdict on the rules. Companies that left the index are missing from every test, which flatters the results.

12 Midas and the Investor

Every result in this report was questioned before it was believed. When something looked wrong, too good, different between indices or against what a long-term investor would say, Midas, who reads

the numbers, put the question to the Investor. The Investor answers only from Warren Buffett's letters to shareholders and names the year. Midas then tested the answer and reported back. Three of the exchanges follow.

Is the dividend rule holding the score back?

Midas: The winners we missed mostly pay no dividend, and they score low on the dividend rules. Should we drop them?

The Investor: (1984) "Berkshire has earned well over market rates on retained earnings, thereby creating over one dollar of market value for every dollar retained. Under such circumstances, any distribution would have been contrary to the financial interest of shareholders, large or small."

Midas: I dropped the two dividend rules from the sixteen-rule score and compared. On the S&P 500 the best ten changed by -0.03 points a year, with a 99% band from -1.05 to +0.41; on the S&P 400 by -0.26. There is no measurable difference. The default now keeps only one dividend rule, the current dividend, and the other is a switch.

Should a cheap price be the test of value?

Midas: Dropping the P/E and price-to-book rules raised the best ten on the S&P 500 but lowered the broad ranking, and the missed winners were growth companies.

The Investor: (1992) "Growth is always a component in the calculation of value." A low ratio of price to book value, a low price-earnings ratio or a high dividend yield are "far from determinative as to whether an investor is indeed buying something for what it is worth", and the opposite characteristics "are in no way inconsistent with a 'value' purchase". Yet "business growth, per se, tells us little about value". (2013) "Price is what you pay, value is what you get."

Midas: I tested a price-to-earnings-growth measure. On all three indices it widened the broad ranking a little and left the best ten slightly lower, within the noise. Used only for companies that pass the quality rule, return on equity, both rose on the S&P 500 and the S&P 400, by 0.3 to 1 point a year. That is small and adds complexity, so it is a switch and not the default.

Is it too good to be true?

Midas: The picks beat the index in all 12 samples that each left out a random tenth of the companies. That looks too good, so I removed the biggest winners as if they had never existed.

The Investor: (1987) "When shortages exist, however, even commodity businesses flourish", and "whenever shortages appear, the typical manager simply can't wait to expand capacity". (1989) "Time is the friend of the wonderful business, the enemy of the mediocre."

Midas: With the three largest winners removed (Applied Materials, Lam Research and Micron) the wallet still beat the index by 3.7, 1.9 and 2.1 points a year for the investors who started in 2016, 2020 and 2023; with five removed, by 0.3 to 2.3 points. The edge is real on the S&P 500 but thin, and it sits in chip and materials companies after 2009. On the S&P 400 it disappears without its three largest winners. Treat it with care.

What this suggests for you. Hold for years, not months. Hold ten companies and not five, since the concentration is the main risk. Treat one or two big winners as luck until they have been tested the same way. Switch rules on or off for the kind of company you want, and read the tests in Section 11 before relying on a number.

13 Where to read more

Table 10: Strategy literature by risk profile.

Profile	Idea	Reading
Defensive	Preserve capital; diversified, cheap, boring; index funds	Graham, The Intelligent Investor (the defensive investor); Bogle, The Little Book of Common Sense Investing
Core	A wonderful company at a fair price, held for years	The Berkshire Hathaway shareholder letters; Dorsey, The Little Book That Builds Wealth
Value with a margin of safety	Buy at a deep discount to value	Graham (the enterprising investor); Klarman, Margin of Safety
Quality growth	Growth at a reasonable price, good management	Fisher, Common Stocks and Uncommon Profits; Lynch, One Up on Wall Street
Systematic value	Rules, ranked	Greenblatt, The Little Book That Beats the Market; Piotroski (2000), the F-score; Novy-Marx (2013), gross profitability
Risk awareness	Cycles and what can go wrong	Marks, The Most Important Thing; Lakonishok, Shleifer and Vishny (1994), contrarian investment

Midas can work from the documents you give it. Download or bring the strategy literature you like, tell Midas the style, and it will answer from those pages and name the source. The Berkshire Hathaway letters are free to read on the company's site. Books are copyrighted: use your own copy. The list below is grouped by the kind of risk each approach accepts. It is a starting point only: the titles are given from general knowledge and have not been checked against the books, so check each one before relying on it.

14 Conclusions

In this sample, the value picks bought twice a year and held are worth more today than the same money in the S&P 500 or in Berkshire Hathaway for each of the four start years. That describes the past under rules and settings chosen while looking at it; it is not a test of the strategy. The lead is uneven: the 8 purchases of Micron Technology (MU) make up 33% of the 2020 investor's value, and the purchases of none trail the index.

Before the strategy is used, three steps are needed: the companies that left the index must be added to the data, the score must be tested on more decisions (each quarter instead of each year), and trading costs and taxes must be included.

A Every purchase

Table 11: Every purchase of the wallet with sells of the investor who started in 2020: weight in the decision's purchase, USD invested (the decision's 100 USD and the proceeds of sales), split-adjusted price at purchase and today, USD sold from it later, USD it keeps today and its total return with dividends, (sold + kept) / invested - 1.

Year	Company	Sector	Weight	In	Bought at	Now	Sold	Kept	Return
2020-01	MU	Technology	12%	12	55.39	884.74		198	+1538%
2020-01	NUE	Materials	12%	12	55.12	273.87		66	+463%
2020-01	CTSH	Technology	11%	11	61.92	65.11		13	+17%
2020-01	SNA	Industrials	10%	10	169.00	355.66		25	+151%
2020-01	FAST	Industrials	9%	9	18.61	51.91		31	+232%
2020-01	TPR	Discretionary	9%	9	26.93	116.29	4	32	+290%
2020-01	AMAT	Technology	9%	9	62.20	390.09		62	+566%
2020-01	PKG	Materials	9%	9	110.72	256.82		26	+183%
2020-01	GILD	Health care	9%	9	65.23	170.68	2	25	+204%
2020-01	LYB	Materials	9%	9	93.86	68.64		10	+12%
2020-07	MU	Technology	12%	12	49.70	884.74		217	+1726%
2020-07	NUE	Materials	11%	11	40.38	273.87		86	+652%
2020-07	CTSH	Technology	11%	11	56.42	65.11		14	+28%
2020-07	HAS	Discretionary	10%	10	74.69	119.09	9	0	-5%
2020-07	HUM	Health care	10%	10	391.55	414.50		11	+12%
2020-07	PKG	Materials	10%	10	99.37	256.82		30	+210%
2020-07	SNA	Industrials	9%	9	134.12	355.66		29	+212%
2020-07	FAST	Industrials	9%	9	21.50	51.91		26	+184%
2020-07	TPR	Discretionary	9%	9	12.97	116.29	9	64	+699%
2020-07	HSIC	Health care	9%	9	58.86	84.88		13	+44%
2021-01	MU	Technology	11%	12	74.05	884.74		142	+1125%
2021-01	NUE	Materials	11%	11	52.51	273.87		64	+468%
2021-01	HUM	Health care	10%	11	412.19	414.50		12	+6%
2021-01	CTSH	Technology	10%	10	79.42	65.11		9	-10%
2021-01	AMAT	Technology	10%	10	86.87	390.09		48	+371%
2021-01	HAS	Discretionary	10%	10	91.96	119.09	7	0	-24%
2021-01	PKG	Materials	10%	10	135.75	256.82		22	+124%
2021-01	LRCX	Technology	9%	10	47.80	239.11		52	+430%
2021-01	GILD	Health care	9%	10	60.11	170.68	3	28	+217%
2021-01	BIIB	Health care	9%	10	242.95	253.62		10	+4%
2021-07	SNA	Industrials	12%	13	223.42	355.66		23	+83%
2021-07	MU	Technology	11%	12	80.11	884.74		138	+1033%
2021-07	HUM	Health care	10%	11	447.63	414.50		11	-2%
2021-07	CMI	Industrials	10%	11	242.13	500.09	0	25	+132%
2021-07	INTC	Technology	10%	11	56.01	65.76	5	3	-19%
2021-07	FAST	Industrials	10%	11	26.03	51.91		24	+127%
2021-07	NUE	Materials	10%	11	96.65	273.87		32	+206%
2021-07	BBY	Discretionary	9%	10	116.37	95.65	2	8	-2%
2021-07	HSIC	Health care	9%	10	75.51	84.88		11	+12%
2021-07	CTSH	Technology	9%	10	69.84	65.11		10	+2%
2022-01	SNA	Industrials	11%	11	213.32	355.66		22	+89%
2022-01	MU	Technology	11%	11	95.75	884.74		102	+845%
2022-01	HUM	Health care	10%	10	465.45	414.50		10	-6%
2022-01	CMI	Industrials	10%	10	219.53	500.09	0	26	+153%
2022-01	HSIC	Health care	10%	10	77.43	84.88		11	+10%
2022-01	INTC	Technology	10%	10	53.21	65.76	5	3	-16%
2022-01	NUE	Materials	10%	10	113.04	273.87		25	+159%
2022-01	AMAT	Technology	10%	10	159.93	390.09		24	+154%
2022-01	PKG	Materials	9%	9	134.61	256.82		20	+119%
2022-01	CTSH	Technology	9%	9	89.57	65.11		7	-21%
2022-07	SNA	Industrials	11%	11	198.24	355.66		23	+101%

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Year	Company	Sector	Weight	In	Bought at	Now	Sold	Kept	Return
2022-07	NUE	Materials	11%	11	105.64	273.87		30	+175%
2022-07	MU	Technology	11%	11	53.65	884.74		179	+1585%
2022-07	AMAT	Technology	10%	10	86.27	390.09		46	+369%
2022-07	SWKS	Technology	10%	10	91.12	87.58		11	+9%
2022-07	GOOG	Communication	10%	10	109.08	278.22		25	+157%
2022-07	GOOGL	Communication	10%	10	108.74	279.84		25	+160%
2022-07	CMI	Industrials	10%	10	194.11	500.09	0	27	+182%
2022-07	INTC	Technology	10%	10	36.34	65.76	7	4	+21%
2022-07	CTSH	Technology	9%	9	68.03	65.11		10	+3%
2023-01	MU	Technology	11%	11	50.37	884.74		196	+1683%
2023-01	SNA	Industrials	10%	11	229.38	355.66		19	+71%
2023-01	GOOG	Communication	10%	11	89.70	278.22		34	+213%
2023-01	GOOGL	Communication	10%	11	89.12	279.84		34	+217%
2023-01	NUE	Materials	10%	11	131.06	273.87		24	+120%
2023-01	CTSH	Technology	10%	11	57.59	65.11		13	+21%
2023-01	LRCX	Technology	10%	10	41.43	239.11		60	+497%
2023-01	SWKS	Technology	10%	10	90.50	87.58		11	+9%
2023-01	CMI	Industrials	9%	10	241.04	500.09	0	22	+124%
2023-01	INTC	Technology	9%	10	26.73	65.76	10	6	+61%
2023-07	STLD	Materials	12%	12	110.78	250.80		29	+137%
2023-07	NUE	Materials	11%	12	165.61	273.87		20	+73%
2023-07	SNA	Industrials	11%	11	286.86	355.66		15	+35%
2023-07	MU	Technology	10%	11	63.90	884.74		149	+1303%
2023-07	CF	Materials	10%	10	70.43	123.30		19	+88%
2023-07	DD	Industrials	10%	10	89.84	140.05		17	+66%
2023-07	CTSH	Technology	10%	10	65.56	65.11		10	+5%
2023-07	LRCX	Technology	9%	10	65.10	239.11		36	+278%
2023-07	SWKS	Technology	9%	10	111.70	87.58		8	-13%
2023-07	GOOG	Communication	9%	9	120.56	278.22		22	+133%
2024-01	STLD	Materials	11%	11	118.91	250.80		25	+119%
2024-01	NUE	Materials	11%	11	175.43	273.87		18	+62%
2024-01	SWKS	Technology	11%	11	109.11	87.58		10	-12%
2024-01	SNA	Industrials	11%	11	288.74	355.66		14	+32%
2024-01	CF	Materials	10%	10	81.61	123.30		15	+61%
2024-01	DD	Industrials	10%	10	97.29	140.05		15	+52%
2024-01	CTSH	Technology	9%	10	74.40	65.11		9	-8%
2024-01	AMAT	Technology	9%	9	154.37	390.09		24	+158%
2024-01	KEYS	Technology	9%	9	155.49	307.20		19	+98%
2024-01	LRCX	Technology	9%	9	74.93	239.11		30	+226%
2024-07	NUE	Materials	11%	12	157.00	273.87		22	+80%
2024-07	SNA	Industrials	11%	11	256.90	355.66		17	+47%
2024-07	STLD	Materials	10%	11	129.15	250.80		23	+100%
2024-07	BG	Staples	10%	11	107.32	96.64		11	-4%
2024-07	SWKS	Technology	10%	11	106.40	87.58		10	-11%
2024-07	EOG	Energy	10%	11	126.54	165.54		15	+42%
2024-07	CTSH	Technology	10%	11	67.74	65.11		11	+0%
2024-07	CF	Materials	9%	10	73.38	123.30		18	+77%
2024-07	ON	Technology	9%	10	69.18	71.67		10	+4%
2024-07	EBAY	Discretionary	9%	10	52.68	121.64	9	10	+101%
2025-01	NUE	Materials	11%	12	114.48	273.87		29	+145%
2025-01	SWKS	Technology	10%	11	88.45	87.58		11	+5%
2025-01	STLD	Materials	10%	11	111.93	250.80		25	+129%
2025-01	BG	Staples	10%	11	77.57	96.64		14	+31%
2025-01	SNA	Industrials	10%	10	335.11	355.66		11	+11%

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Year	Company	Sector	Weight	In	Bought at	Now	Sold	Kept	Return
2025-01	EOG	Energy	10%	10	124.71	165.54		15	+42%
2025-01	CTSH	Technology	10%	10	76.41	65.11		9	-12%
2025-01	AMAT	Technology	9%	10	163.87	390.09		24	+141%
2025-01	ON	Technology	9%	10	61.71	71.67		11	+16%
2025-01	CF	Materials	9%	10	85.83	123.30		14	+49%
2025-07	ON	Technology	11%	13	53.60	71.67		17	+34%
2025-07	SWKS	Technology	11%	13	77.16	87.58		15	+18%
2025-07	SNA	Industrials	11%	13	317.86	355.66		15	+16%
2025-07	EW	Health care	10%	12	77.18	92.01		14	+19%
2025-07	BG	Staples	10%	11	80.25	96.64		14	+25%
2025-07	FSLR	Technology	10%	11	162.96	174.40		12	+7%
2025-07	CTSH	Technology	10%	11	80.44	65.11		9	-17%
2025-07	AMAT	Technology	9%	11	183.76	390.09		23	+114%
2025-07	STLD	Materials	9%	11	130.16	250.80		21	+95%
2025-07	EOG	Energy	9%	11	121.19	165.54		15	+43%
2026-01	ON	Technology	11%	13	56.70	71.67		16	+26%
2026-01	SNA	Industrials	11%	12	350.95	355.66		13	+3%
2026-01	SWKS	Technology	10%	12	64.40	87.58		16	+39%
2026-01	EW	Health care	10%	12	85.33	92.01		13	+8%
2026-01	CPRT	Industrials	10%	11	37.77	30.13		9	-20%
2026-01	CTSH	Technology	10%	11	81.28	65.11		9	-18%
2026-01	BG	Staples	10%	11	92.64	96.64		12	+6%
2026-01	EOG	Energy	10%	11	107.27	165.54		17	+59%
2026-01	NUE	Materials	10%	11	169.40	273.87		18	+63%
2026-01	STLD	Materials	9%	10	176.06	250.80		15	+44%
2026-07	SNA	Industrials	11%	12	403.28	355.66		11	-11%
2026-07	CTSH	Technology	11%	12	41.07	65.11		19	+59%
2026-07	FSLR	Technology	11%	11	232.08	174.40		9	-25%
2026-07	SWKS	Technology	10%	11	65.93	87.58		15	+33%
2026-07	RMD	Health care	10%	11	201.10	229.94		13	+15%
2026-07	EW	Health care	10%	11	91.99	92.01		11	+0%
2026-07	BF-B	Staples	10%	11	25.83	28.71		12	+12%
2026-07	PODD	Health care	9%	10	159.79	149.68		9	-6%
2026-07	CF	Materials	9%	10	108.16	123.30		11	+15%
2026-07	HUM	Health care	9%	9	409.42	414.50		9	+1%

B Wallet contents by year

Table 12: Wallet with sells of the investor who started in 2020: value of each holding after each January's trades and today (USD), and the money put in and the proceeds of sales to date, July decisions included.

Holding	2020	2021	2022	2023	2024	2025	2026	Now
AMAT	9	23	53	44	80	95	173	252
SNA	10	22	53	82	128	178	218	236
PKG	9	35	45	45	59	83	80	99
NUE	12	38	105	149	225	170	266	434
MU	12	45	84	66	122	130	471	1,321
LYB	9	9	10	10	12	9	6	10
GILD	9	18	23	24	24	28	37	53
FAST	9	23	43	34	47	53	62	81
TPR	9	31	30	29	30	53	105	96
CTSH	11	40	67	63	103	130	163	152
HUM		21	46	50	48	26	27	53
HSIC		10	32	32	31	28	31	35
HAS		23	25	16	9	1	1	1
BIIB		10	10	11	11	6	7	10
LRCX		10	15	19	55	53	138	178
INTC			20	28	51	22	12	17
BBY			9	8	8	7	5	8
CMI			20	45	46	67	103	100
GOOGL				19	29	40	67	59
GOOG				19	40	55	91	81
SWKS				20	45	58	66	107
CF					21	45	43	78
STLD					25	44	96	138
DD					21	20	27	31
KEYS					9	10	12	19
EOG						21	39	62
BG						19	48	51
ON						19	43	55
EBAY						12	10	10
FSLR							19	20
EW							25	38
CPRT							11	9
BF-B								12
PODD								9
RMD								13
Put in	100	300	500	700	900	1,100	1,300	1,400
Sold	0	4	13	18	23	36	66	74
Total	100	358	691	809	1,276	1,480	2,503	3,927

C Testing the weighting

Table 13: Each rule's weight w_r on the picks' dates (overall, for the picks that beat the S&P 500 and for the others), the dollar-weighted mean points, the gap and the rank correlation with the return a year above the S&P 500.

Rule	w_r	w_r beat	w_r not	Pts beat	Pts not	Gap	Corr.
NN	1.00	1.00	1.00	3.59	2.69	+0.90	-0.03
G7	0.91	0.91	0.91	3.67	3.54	+0.14	+0.13
G6	0.88	0.87	0.88	3.99	3.80	+0.19	+0.14
B1	0.84	0.84	0.84	3.24	3.56	-0.32	-0.09
G2b	0.83	0.83	0.83	4.86	4.68	+0.18	-0.00
E2	0.82	0.81	0.82	4.86	4.68	+0.18	-0.00
G2a	0.76	0.76	0.76	4.88	4.54	+0.35	+0.26
E1	0.59	0.59	0.60	4.88	4.54	+0.35	+0.26
G5	0.53	0.53	0.53	3.85	3.37	+0.48	+0.20
E4	0.19	0.19	0.19	2.04	2.29	-0.25	+0.01

Each rule's points are multiplied by its weight, one less the share of companies passing it on the date, so a rule few companies pass counts more. A good weighting gives most weight to the rules whose points went with better returns. Table 13 sets each rule's mean weight on the dates the picks were made against two measures of how well it told the 220 picks apart: the gap between the mean points of the picks that beat the S&P 500 and of those that did not, weighted by dollars as in Figure 4, and the rank correlation of the rule's points with each pick's return a year above the S&P 500. Figure 9 plots weight against gap; a good weighting puts its rules on a rising line, with the heavy rules above zero.

Across the rules, the rank correlation of weight and gap is +0.02, and of weight and correlation -0.48.

The weighting is unrelated to which rules worked. Weighted heavily but with a negative gap: return on equity (B1). A positive gap but a light weight: current ratio (G2a), 6-year earnings growth (G5), current ratio (lenient) (E1) and debt to working capital (lenient) (E2). Any change to the weights must be tested on years it was not fitted on.

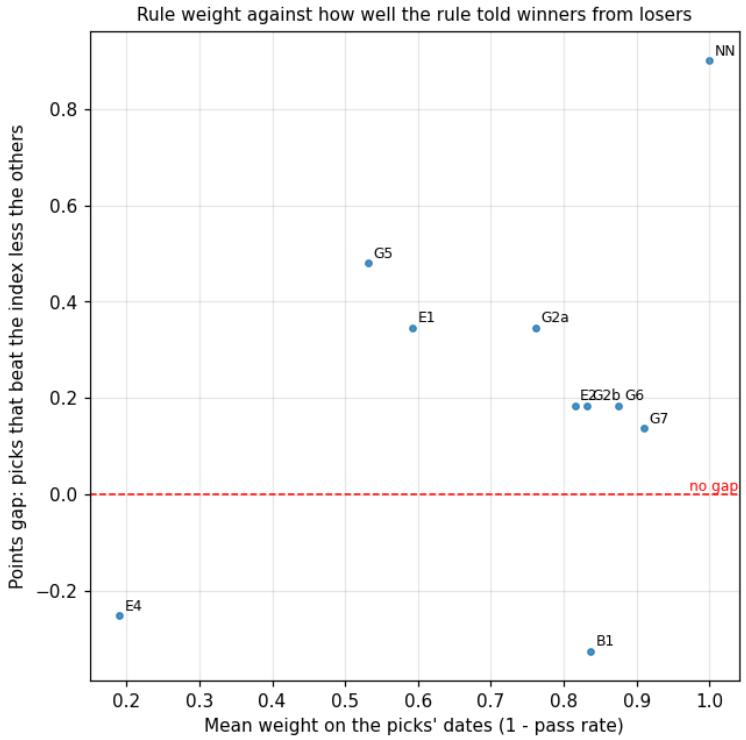


Figure 9: Each rule's mean weight against the gap in points between the picks that beat the S&P 500 and the others, weighted by dollars; rules above the line told winners from losers.

D Forward information

Table 14: Truncation check: companies scored on the date, companies whose score differs when every input is cut at the date (per score), and whether the picks are the same.

Date	Companies	Score	Score 0 to 1	Unweighted	Share passed	Same picks
2018-12-21	296	0	0	0	0	yes
2022-12-16	339	0	0	0	0	yes
2025-12-19	376	0	0	0	0	yes

A score must use only what was known on its date. Four guards are built in. A company's figures count from the day its annual report was filed with the SEC, and of all the figures filed for a year only the first reported is used, never a later restatement. Prices are the close on the scoring date and dividends are counted by their ex-date. Each January and July purchase uses the scores of the rebalance date two weeks before it. Rule weights come from the companies on the date alone.

These guards are tested by cutting every input at a date as if it were that day (filings filed on or before it, prices and dividends up to it), scoring the companies again and comparing the scores with the full run (Table 14). Any difference means information from after the date reached a score. On 21 December 2018, 16 December 2022 and 19 December 2025 the picks were the same, and the scores were identical on 21 December 2018, 16 December 2022 and 19 December 2025. The same check runs as a unit test. A rule whose window reaches back further than a company's price history (the 20-year dividend record of a younger company) is not evaluated rather than failed.

Three sources of forward information are outside this test. Only today's S&P 500 members are scored. Each company counts from the date it entered the index, as the index's list gives it. Companies that later left the index are missing, which flatters every result. Sectors are today's, used to leave out Financials and Real Estate in earlier years. And the score's design and settings

were chosen while looking at 2016 to 2026, so no period tests it fairly: 2022 onwards, once meant as a held-out test, was used to choose between models.

D.1 Companies missing from the data

Table 15: Companies that left the S&P 500 since each start year, all missing from the data: in all, on their market value, and through an acquisition or another reason.

Start	Left the index	On market value	Acquired or other
2016	230	97	133
2020	125	75	50
2023	69	47	22
2025	37	24	13

Table 15 counts the companies that left the S&P 500 after each start year; none of them is in the data, of any sector. Most left through an acquisition, often at a premium; those that dropped out on their market value are the likely losers a real investor could have picked. The later the start, the fewer are missing, which is why the investor who started in 2020 is the reference of this report. The count starts with the index's list of changes, which begins in July 1976: removals before that are not counted, so for earlier start years it understates the gap. The bias is removed only when the historical members of the index are added to the data.

E The rules

Table 16: The rules: what each measures, the better direction and when a company passes.

Rule	Name	What it measures	Better	Passes when
G2a	current ratio	Current assets divided by current liabilities: what the company owns against what it owes within a year.	higher	at least 2.0
G2b	debt to working capital	Long-term debt divided by net current assets (current assets less current liabilities).	lower	at most 1.0
G5	6-year earnings growth	Average earnings per share of the last 3 years over the average of the 3 years before them, within a window of 6 years.	higher	at least 1.33 times the earlier average
G6	P/E	Price over the average earnings per share of the last 3 years.	lower	at most 15.0
G7	P/E times P/B	P/E as in G6 times the price over book value per share.	lower	P/B at most 1.5, or the product at most 22.5
E1	current ratio (lenient)	Current ratio, as G2a.	higher	at least 1.5
E2	debt to working capital (lenient)	Long-term debt over net current assets, as G2b.	lower	at most 1.1
E4	current dividend	Dividends paid over the last 365 days, as a yield on the price.	higher	any dividend
NN	net-net	Market value over net current asset value (current assets less all liabilities).	lower	at most 0.67
B1	return on equity	Number of the last 3 years with return on equity (net income over equity) of at least 15%.	higher	at least 2 of those years, and long-term debt at most 0.5 times equity

Table 16 explains each rule: what it measures, whether a higher or a lower value earns more points, and when a company passes it. The points come from the company's rank on the measure among all companies on the date; the pass threshold sets the rule's weight. Rules G follow Graham's defensive investor (The Intelligent Investor, chapter 14), rules E his enterprising investor (chapter 15) and his net-net test (NN), B1 Buffett's return on equity, and NI1 was added for net income growth. Two pairs test the same measure at different strictness (G2a and E1, G2b and E2), so the current ratio and debt count twice.